What Happens in Paris Does Not Stay in Paris: Trade Fairs and Search and Matching Frictions

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THE BROADER PROJECT: GLOBALIZATION AND INDUSTRIAL POLICY

- **Globalization and Industrial Policy in the Austro-Hungarian Monarchy**
  - Joint project by CEU (Vienna) and LMU (Munich), two teams
- **Core question:** What are the mechanisms and effectiveness of industrial policy tools commonly used today?
- **Approach:** Study these tools in historical context exploit identification opportunities

WHY THE AUSTRO-HUNGARIAN MONARCHY (AHM)?

GDP per capita: Austria-Hungary relative to Germany, and Hungary relative to Austria

	1870	1890	1913
Austria-Hungary as % of Germany	66	60	54
Hungary as % of Austria	68	77	77

Source: Computed from Klein, Schulze & Vonyó (2017, Table 5.1), based on Schulze (2000, 2007b). GDP per capita in 1990 Geary-Khamis dollars. Austria-Hungary composite uses approximate population weights 58-42

WHY THE AUSTRO-HUNGARIAN MONARCHY (AHM)?

- **Relevant for today's policy debates**
  - Middle-income country undergoing industrialization
  - Active use of industrial policy tools still common today
  - Multi-ethnic context relevant for diverse economies
- **Clean setting for identification**
  - Limited communication: postal mail, costly travel
  - Fewer overlapping policies compared to modern contexts
  - Government actions more isolated and observable
- **Under-studied setting**
  - Literature focused on US, UK, France, Germany
  - Little micro-level evidence from AHM

PROJECTS

1. **Exhibitions as export promotion** $\leftarrow$ *today's talk*
   - 1900 Paris World Exhibition as a natural experiment
   - Effect on firm exports and growth
2. **Export-contingent input tariff reductions**
   - Milling industry, 1885–1900
   - How reduced input costs affect technology adoption and location choice
3. **Catalog of industrial policies in AHM**
   - Comprehensive review of all IP tools used, 1870–1910
4. **Micro data catalog**
   - Digitalization, Matching of micro datasets. Public by end.

DATA COLLECTION: A MAJOR INVESTMENT

- **Novel firm-level panel data** from previously unexplored archival sources
  - Historical catalogs, government surveys, export directories
  - Period: 1894–1914
- **Digitization pipeline**
  - Scan $\rightarrow$ OCR (Google Document AI) $\rightarrow$ AI-assisted parsing $\rightarrow$ manual validation
  - Challenges: Hungarian language, historical orthography, unstructured data
- **Entity resolution**
  - Link firms across 5+ independent datasets
  - Fuzzy name matching, address standardization, extensive manual checking
- **Result:** First comprehensive micro-panel for AHM industrial firms

Today's talk:

Trade Fairs and Search and Matching Frictions

Evidence from the 1900 Paris World Exhibition

- Motivation
- Data, Data, Data and History
- Identification and econometrics
- Results and economics

Introduction

MOTIVATION

- **What prevents firms from exporting?**
  - Tariffs?
  - But: Even when tariffs were low (pre–trade war), export participation was below what trade theory predicts ("dark trade costs", Head and Mayer 2013)

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- **Matching frictions between buyers and sellers?**
  - Importer-exporter links relevant for export entry, export growth, access to inputs, firm-to-firm networks, transmission of shocks
  - Various approaches to reduce search and matching frictions: export promotion agencies, referrals, transport/communication infrastructure (RCT, observational)

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- **Common way for firms to find trading partners: trade fairs**
  - Difficult to evaluate due to selection bias and hard to isolate policy effects from firms' unobserved search efforts
  - This paper: use 1900 Paris world fair to estimate its effect on export outcomes and firm development

PAPER IN A NUTSHELL

- **Research question: Does technology fair lead to more exporting?**
  - Selection bias: firms that participate often intend to export anyways, are more successful
  - Historical context: limited private search efforts, as communication was costly and slow

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- **Findings**
  - Confirm positive selection
  - Positive upper and lower bounds for treatment effect on exporting
  - Smaller if more Hungarian competitors present, larger if more international buyers present

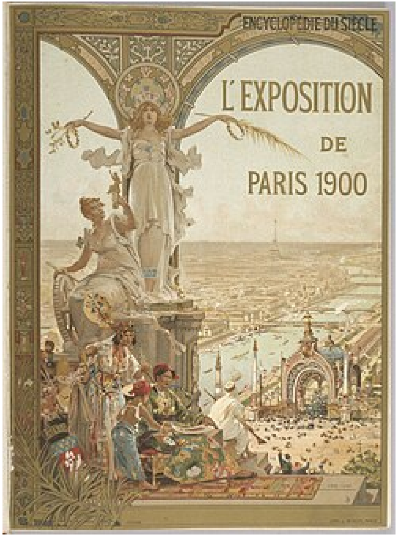
LITERATURE

- **International firm-to-firm trading patterns**
  - Who matches with whom: Rauch 2001; Rauch and Trindade 2002; Sugita et al. 2023; Bernard and Jensen 2004; Eaton et al. 2022; Benguria 2021; Demir et al. 2024; Arkolakis et al. 2025; Miyauchi 2024
  - Reducing search frictions: Atkin et al. 2017; Jensen 2007; Jensen and Miller 2018; Bernard et al. 2019; Cai et al. 2024; Verhoogen 2008; Allen 2014
- **Export promotion policies**
  - Buus et al. 2025; Munch and Schaur 2018; Carballo and Volpe Martincus 2008; Martincus and Carballo 2010; Broocks and Van Biesebroeck 2017; Munch et al. forthcoming; Van Biesebroeck et al. 2015; Hiller 2012; Görg et al. 2008; Schminke and Van Biesebroeck 2013
- **Industrial policies in the 19th century**
  - Juhász and Steinwender 2018; Juhász 2018; Donaldson and Hornbeck 2016; Romero 2023; Moser 2012, 2005; Hanlon 2020; Alfaro et al. 2022
- **Partial identification**
  - Manski 2003; Lee 2009; Manski and Pepper 2000; Tamer 2010; Ho and Rosen 2017; Blundell et al. 2007

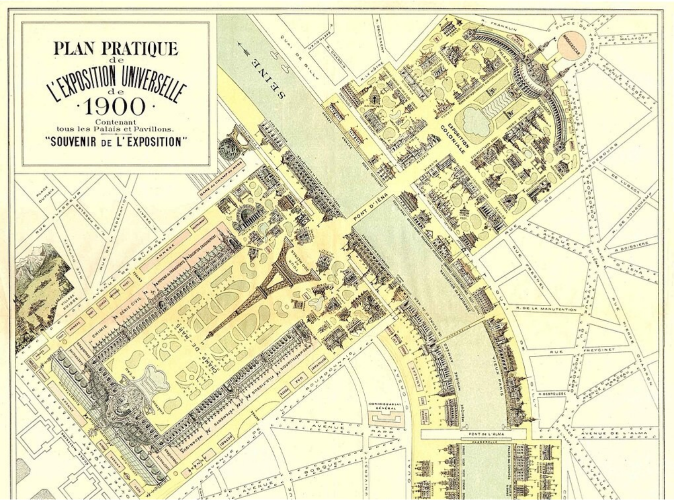
The 1900 Paris exhibition

1900 PARIS EXPOSITION UNIVERSELLE

- **Among largest international commercial gatherings in history**
  - 48 million visitors
  - 70,000 firms
  - 56 countries
  - Innovations: diesel engine, trolleybus...
- **Trade fair**
  - Firms showcase their products and innovations
  - Firms put price tags on products
  - Networking: meetings, lunches, dinners, events
- **Industry pavilions**
  - Diesel engine, moving sidewalk, motion picture, passenger trolleybus



PARIS EXPO (COUNTRIES) + TRADE FAIR (INDUSTRIES)



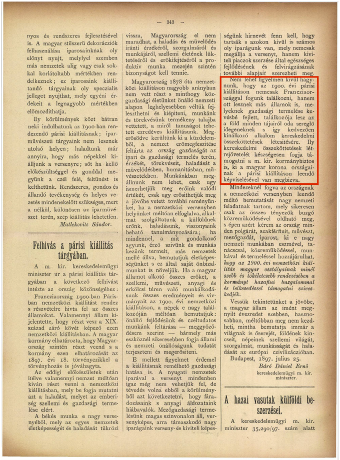
HUNGARY AT THE PARIS EXHIBITION

- **Hungarian Kingdom**
  - Part of Austro-Hungarian Monarchy (AHM), 1867–1918
  - Great deal of independence by 1890
  - Own finances, economic and industrial policies
  - But: common foreign policy, customs union
- **Hungary at the Paris exhibition**
  - Mostly agriculture, wants to industrialize more
  - Used trade fair as export promotion policy
  - First time as independent participant
  - Sent 3,000 exhibitors (5th largest country)

GOVERNMENT: TRADE FAIR AS EXPORT PROMOTION POLICY

"… the 1900 Paris Exhibition … **will be a meeting place for foreigners flocking there from all parts of the world, and thus a favorable opportunity for establishing commercial connections.** The Hungarian Royal Commissioner, who … is … organizing the representation of the countries of the Hungarian Crown … will support the establishment of such commercial connections."

Hungarian Industrial Journal 1897 ed. 2, by Baron Ernő Dániel, Hungarian Royal Minister of Commerce



— 343 —

Nem lehet figyelmen kívül hagynunk, hogy az 1900. évi párisi kiállításon nemcsak Francziaországgal fogunk találkozni, hanem ott lesznek más államok is, melyeknek gazdasági termelése kevésbé fejlett, találkozója lesz az a föld minden tájairól oda sereglő idegeneknek s így kedvezően kínálkozó alkalom kereskedelmi összeköttetések létesítésére. Ily kereskedelmi összeköttetések létrejövetelét készségesen fogja támogatni a m. kir. kormánybiztos is, ki a magyar korona országainak a párisi kiállításon leendő képviselésével van megbízva.

Felhívás a párisi kiállítás tárgyában.

Báró Dániel Ernő
kereskedelemügyi m. kir. miniszter.

A hazai vasutak külföldi beszerzései.

HUNGARY'S PREPARATIONS FOR 1900 PARIS

- **Preparations started in 1887(!)**
  - Selection committee
  - Budapest Millennial Exhibition in 1896 as a "trial exhibition"

- **Budapest Millennial Exhibition in 1896**
  - Like a 'census' of Hungarian firms $\Rightarrow$ over 9,000 exhibitors
  - Independent juries gave awards for each exhibit's export potential
    - 34 product group juries, 480 members; half elected/half appointed; formal regulations and procedures
  - Four award categories: gold, silver, bronze, honorable mention
    - Majority vote of group jury; jury members could not get prizes

GOVERNMENT: TRADE FAIR AS EXPORT PROMOTION POLICY

- **Policy**
  - Selection
  - Subsidized transport costs for Paris exhibitors
- **Constraints:**, government did not select only the best for Paris
  - Space limitations in Paris
  - Aim of industrial representation
- **Consequences** (key for identification):
  - Not representative industry portfolio
  - Good but not just best firms

Empirical strategy

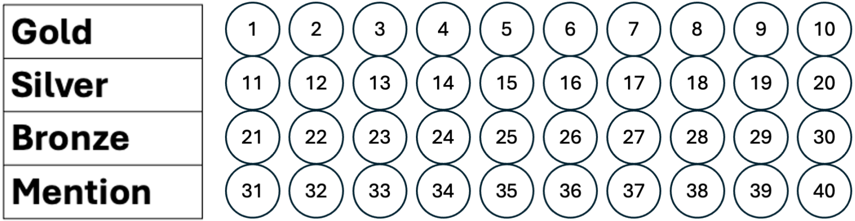
INTUITION

- **Identification strategy: estimating causal bands**
  - Use trial-exhibition to get a proxy for unobserved firm export potential: award categories
  - Even within award categories, probably positive selection
  - Idea: compare untreated firms in better award category to treated firms in lower award category $\Rightarrow$ should downward bias results, provides causal bands

- **Simple example**
  - There are four award categories, each with 10 firms
  - The government will select firms with the top export potential for Paris

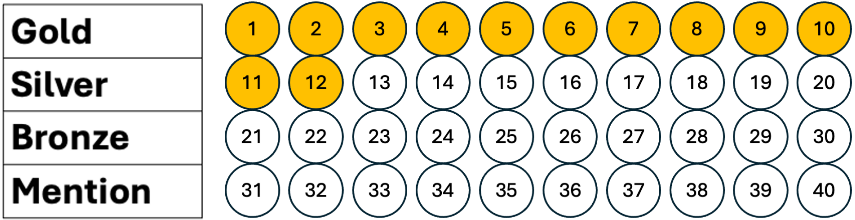
  - We show how we use this to identify causal bounds

GIVE SIMPLIFIED EXAMPLE FOR INTUITION



- Assume that numbers represent unobserved export potential
- We only observe bins of export potential = awards

GOVERNMENT SELECTS FIRMS WITH HIGH EXPORT POTENTIAL

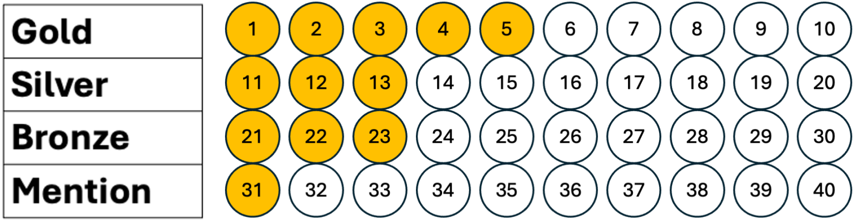


- Space constraints: suppose there are 12 places for Hungary in the 1900 Paris exhibition
- Positive selection

IS INCONSISTENT WITH INDUSTRIAL REPRESENTATION

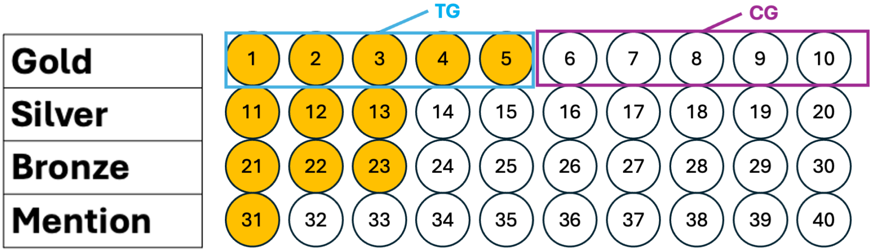
- Two industries: food (green) and textiles (red)
- If some industries have better export potential, they would be overrepresented
    - Example: food industry (green) had advanced milling technology

GOVERNMENT MUST SELECT FIRMS IN LOWER CLASSES



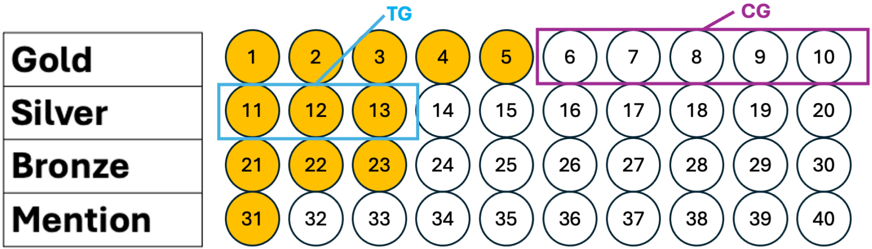
- As a result, to ensure industrial representation and other constraints, government also selected firms in lower award classes, i.e., with smaller export potential

UPPER BOUND: TE WITHIN AWARD CATEGORIES



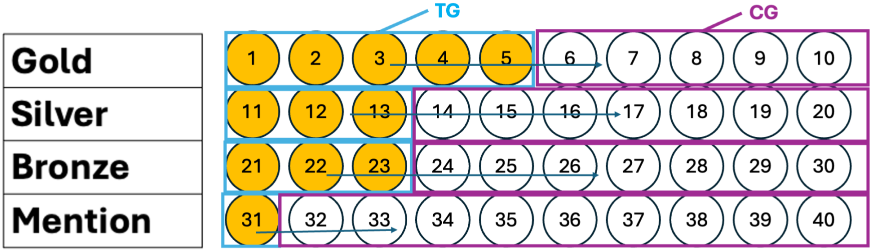
- Reasonable to assume that they selected the best firms within an award group ⇒ positive selection
- Comparing firms within an award category: upwards bias = upper bound

LOWER BOUND: TE RELATIVE TO “BETTER” CONTROL GROUP



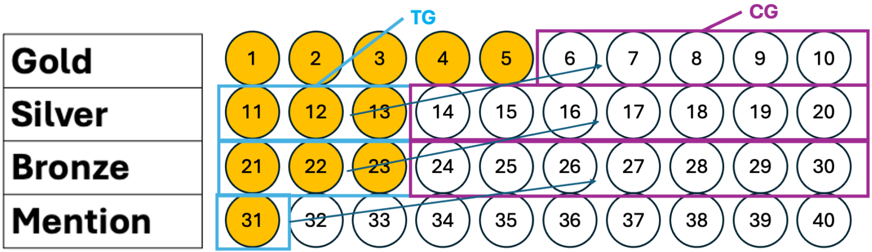
- Compare treated firms from lower award class to untreated firms from higher award class: downwards bias = lower bound

STACKING COMPARISONS: UPPER BOUND



- “Upper bound sample”: all firms
- Award group dummies to ensure comparison within award groups

STACKING COMPARISONS: LOWER BOUND



- “Lower bound sample”: drop treated firms from top category, and untreated firms from lowest category

STACKING

- Stack across award/comparison groups
  - To get enough power
  - Dummy variables for each stacked comparison group
  - Assume poolability

ESTIMATING EQUATIONS FOR UPPER AND LOWER BOUND

$$\Delta y_i = \beta \, \text{exhibitor}_i + \text{BoundGroupFE}_j + FE_p + y_{i,pre} + \epsilon_i$$

- **exhibitor$_i$**: dummy if attending Paris exhibition in 1900
- **Upper bound for β:**
  - $BoundGroupFE_j$ = dummies group untreated firms in <u>same</u> award class as treated firms
  - Sample: all firms
- **Lower bound for β:**
  - $BoundGroupFE_j$ = dummies group untreated firms in <u>higher</u> award class vs treated firms
  - Sample: only firms that belong to a lower bound group (drops treated firms in top award class and untreated firms in lowest award class)

IDENTIFYING ASSUMPTION 1: WITHIN-AWARD POSITIVE SELECTION

Assumption 1 (upper bound, within-award selection)

$$E[Y(0) \mid D = 1, A = a] \; \geq \; E[Y(0) \mid D = 0, A = a] \quad \forall a$$

This implies that the within-award comparison is an upper bound on the ATT:

$$\begin{aligned}
\beta^a_{\text{upper}} &= E[Y(1) \mid D = 1, A = a] - E[Y(0) \mid D = 0, A = a] \\
&\geq E[Y(1) \mid D = 1, A = a] - E[Y(0) \mid D = 1, A = a] = \tau^a_{\text{ATT}}
\end{aligned}$$

- Selection probability drops monotonically with award quality (60% $\rightarrow$ 7%)
- Plausible that even within award bins, better firms are more likely selected
- Similar to standard positive selection assumption in program evaluation

IDENTIFYING ASSUMPTION 2: CROSS-AWARD ORDERING

Assumption 2 (lower bound, cross-award ordering)

$$E[Y(0) \mid D = 0, A = a - 1] \;\geq\; E[Y(0) \mid D = 1, A = a] \quad \forall a$$

This implies that the across-award comparison is a lower bound on the ATT:

$$\begin{aligned} \beta^a_{\text{lower}} &= E[Y(1) \mid D = 1, A = a] - E[Y(0) \mid D = 0, A = a - 1] \\ &\leq E[Y(1) \mid D = 1, A = a] - E[Y(0) \mid D = 1, A = a] = \tau^a_{\text{ATT}} \end{aligned}$$

- Untreated firms from a higher award class have higher export potential than treated firms from the next lower class
- Stronger assumption, but natural given discrete award bins
- The “worst” gold medalist who didn’t go to Paris plausibly has higher export potential than the “best” silver medalist who did

ADVANTAGE OF THESE ASSUMPTIONS

- **Key advantage:** We don't need unconfoundedness, only monotonicity of selection

VALIDATION TEST: UPPER VS LOWER BOUND

Lemma (validation test)

Under Assumptions 1 and 2, the upper bound estimate must be weakly larger than the lower-bound estimate:

$$\beta_{\text{upper}} \geq \beta_{\text{lower}}$$

Interpretation:

- If $\beta_{\text{upper}} < \beta_{\text{lower}}$, at least one of Assumptions 1 or 2 is violated
- If $\beta_{\text{upper}} \geq \beta_{\text{lower}}$, the assumptions pass this necessary test, but this is not sufficient (one or both assumptions could still be violated)

BENEFITS OF CAUSAL BOUNDS VS. ALTERNATIVE APPROACHES

- **Matching estimators** (CEM, PSM, nearest neighbor based on awards)
  - Compare treated and untreated firms within the same award category
  - $=$ our upper bound: likely subject to residual positive selection bias
  - Matching only uses “close” matches
- **Our approach**
  - Exploits additional information: “close from above” vs. “close from below”
  - Bounds the treatment effect rather than relying on untestable assumptions
  - Built-in validation test ($\beta_{\text{upper}} \geq \beta_{\text{lower}}$)
- **Fuzzy RDD analogy**
  - Award categories $\approx$ binned running variable
  - But: binning masks selection within categories $\Rightarrow$ standard fuzzy RDD invalid
  - Our approach handles this explicitly through bounding

RELATION TO PARTIAL IDENTIFICATION LITERATURE

- **Classic Manski bounds:** very wide under minimal assumptions ($Y \in [0, 1]$)
- **Tightening approaches:**
  - Monotone Treatment Response (MTR): $Y(1) \geq Y(0)$
  - Monotone Treatment Selection (MTS): treated units have weakly higher potential outcomes (Manski and Pepper, 2000)
  - Trimming bounds (Lee, 2009)
- **Our innovation:**
  - Exploit discrete award categories to construct *two* distinct control groups
  - Transparent economic intuition: “similar” vs. “superior” export potential
  - Testable implication: upper $\geq$ lower
  - Leverages unique institutional feature (trial exhibition + government constraints)
- We compare our bounds to Manski bounds in the results section

Data sources and digitization

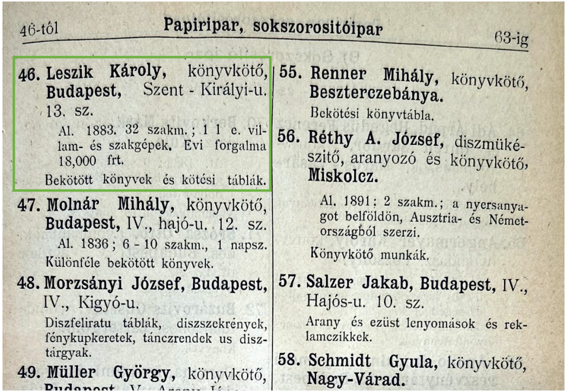
HISTORICAL "PANEL DATA" SET FOR HUNGARY 1896–1905

Combine data from several sources:

1. **Budapest Millenial Exhibition catalog, 1896**
   - Firm-level variables: export status, workers, horse power, output
2. **Budapest Millenial Exhibition Brochure, 1896**
   - Awards
3. **Paris Exposition Universelle catalog, 1900**
   - Treatment: exhibitors
   - Number of competitors and potential buyers
4. **Firm survey by the Hungarian Museum of Commerce, 1898 and 1904**
   - Outcomes: export capability (incl. Austria) ▸ more, workers, horsepower
5. **Export Compass by the Austrian Ministry of Commerce, 1905**
   - Outcomes: export status (outside Austria-Hungary)
6. **Corporate Registry (Hungary), 1894-1904**
   - Firm birth, exit, transformation

BUDAPEST MILLENIAL EXHIBITION CATALOG, 1896

- Number of workers (including skilled/unskilled)
- Capacity of machinery in horsepower (hp)
- Products offered and exhibited
- Special products and patents
- Output measured in Hungarian forints
- Export destinations
- Year of firm establishment
- Exhibition category (industry)


46-tól — Papiripar, sokszorositóipar — 63-ig

46. Leszik Károly, könyvkötő, **Budapest,** Szent - Királyi-u. 13. sz.

Al. 1883. 32 szakm.; 1 1 e. villam- és szakgépek. Evi forgalma 18,000 frt.

Bekötött könyvek és kötési táblák.

47. Molnár Mihály, könyvkötő, **Budapest,** IV., hajó-u. 12. sz.

Al. 1836; 6 - 10 szakm., 1 napsz.

Különféle bekötött könyvek.

48. Morzsányi József, Budapest, IV., Kigyó-u.

Diszfeliratu táblák, diszszekrényck, fénykupkeretek, tánczrendek és dísztárgyak.

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55. Renner Mihály, könyvkötő, **Beszterczebánya.**

Bekötési könyvtábla.

56. Réthy A. József, diszmüké-szitő, aranyozó és könyvkötő, **Miskolcz.**

Al. 1891; 2 szakm.; a nyersanyagot belföldön, Ausztria- és Németországból szerzi.

Könyvkötő munkák.

57. Salzer Jakab, Budapest, IV., Hajós-u. 10. sz.

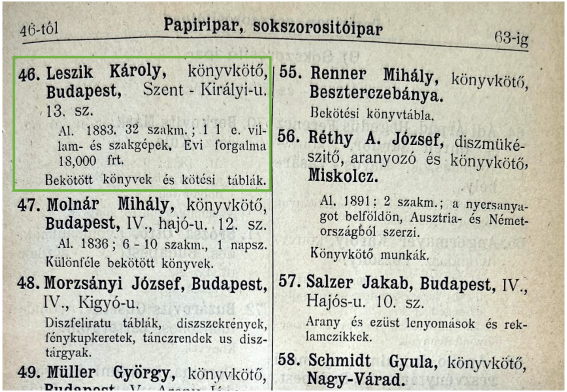
Arany és ezüst lenyomások és reklamczikkek.

58. Schmidt Gyula, könyvkötő, **Nagy-Várad.**

DATA EXTRACTION STEPS, BUDAPEST CATALOG 1896

Digitization steps

- Source: *General catalog of the 1896 Millennium National Exhibition*
- Scan, OCR, and regex
- Challenge: unstructured data, manual separation of fields were required for most observations
- Awards are processed similarly, matched to exhibitors by their names — fuzzy matching



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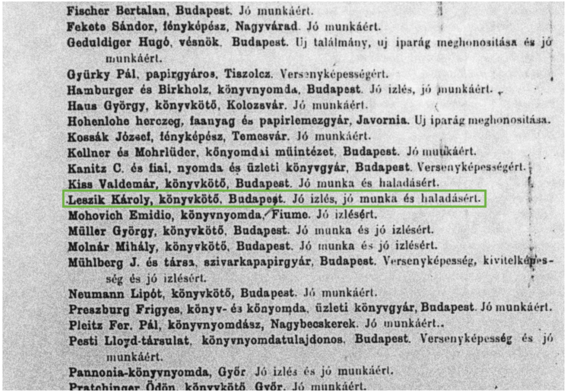
back

BUDAPEST MILLENIAL EXHIBITION BROCHURE, 1896

- Name of awardee
- Address
- Product category
- Award category (class 1–4, 1 is “Gold”)
- Explanation for the award



Fischer Bertalan, Budapest. Jó munkáért.
Fekete Sándor, fényképész, Nagyvárad. Jó munkáért.
Geduldiger Hugó, vésnök. Budapest. Uj találmány, uj iparág meghonositása és jó munkáért.
Gyürky Pál, papirgyáros, Tiszolcz. Versenyképességért.
Hamburger és Birkholz, könyvnyomda, Budapest. Jó izlés, jó munkáért.
Haus György, könyvkötő, Kolozsvár. Jó munkáért.
Hohenlohe herczeg, faanyag és papirlemezgyár, Javornia. Uj iparág meghonositása.
Kossák József, fényképész, Temesvár. Jó munkáért.
Kollner és Mohrlüder, könyomdai müintézet, Budapest. Jó munkáért.
Kanitz C. és fiai, nyomda és üzleti könyvgyár, Budapest. Versenyképességért.
Kiss Valdemár, könyvkötő, Budapest. Jó munka és haladásért.
Leszik Károly, könyvkötő, Budapest. Jó izlés, jó munka és haladásért.
Mohovich Emidio, könyvnyomda, Fiume. Jó izlésért.
Müller György, könyvkötő, Budapest. Jó munka és jó izlésért.
Molnár Mihály, könyvkötő, Budapest. Jó munka és jó izlésért.
Mühlberg J. és társa, szivarkapapirgyár, Budapest. Versenyképesség, kivitelképesség és jó izlésért.
Neumann Lipót, könyvkötő, Budapest. Jó munkáért.
Preszburg Frigyes, könyv- és könyomda, üzleti könyvgyár, Budapest. Jó munkáért.
Pleitz Fer. Pál, könyvnyomdász, Nagybecskerek. Jó munkáért.
Pesti Lloyd-társulat, könyvnyomdatulajdonos, Budapest. Versenyképesség és jó munkáért.
Pannonia-könyvnyomda, Győr. Jó izlés és jó munkáért.
Pratchinger Ödön, könyvkötő, Győr. Jó munkáért.

PARIS CATALOG: EXHIBITING FIRMS ARE TREATED

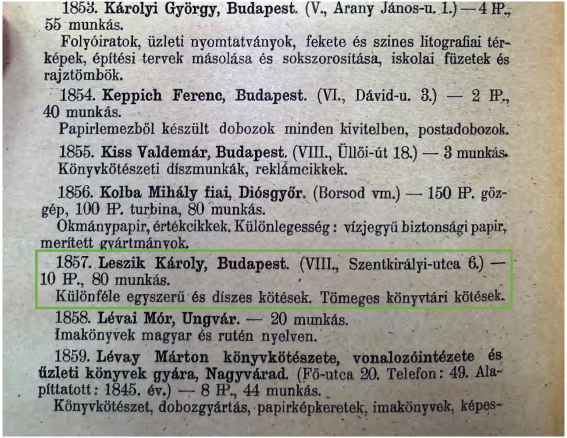
		Gottermayer Nándor könyvkötészet — Buchbindeanstalt ...	Budapest IV, Királyi Pál utca 5.
9	1459	Riedl Ödön (Edmund) fésüs — Kammacher	Arad
10	1462	Táfler Jakab első orosházi olaszcziroksepr ő — I. italienische Kehrbesen- und Bürstenfabrik	Orosháza
11	1472	Kratky Antal fésüsmester — Kammachermeister	Budapest VIII, Népszinház-utca 13.
12	1652	Gröber Lajos	Budapest IV, Kigyó-u. 6.
13	1683	Burg Ármin bőrdiszműárugyáros — Ledergalanteriewaarenfabrik	Budapest V, Váci-körut 6.
14	1684	Schnitzer Jakab könyvkötő és diszműkészitő — Buchbinder und Galanteriewaarenerzeuger	Maros-Vásárhely
15	1720	Krausz Károly és József	Budapest VI, Andrássy-u. 43.
16	1980	Leszik Károly könyvkötő — Buchbinder	Budapest VIII, Szentkirályi-u.
17	164	Torday József	B.-Hunyad

DIGITIZATION PIPELINE: PARIS EXHIBITION

- **Step 1: Scanning**
  - Adobe Scan for high-quality images from archival microfilm
- **Step 2: OCR**
  - Google Document AI — works efficiently with Hungarian language
  - Trained model to separate multi-column pages
- **Step 3: Parsing**
  - AI tools to separate exhibitors from each other
  - Regular expressions to extract structured fields (workers, HP, address)
  - Challenge: unstructured data, manual separation needed for most observations
- **Step 4: Validation**
  - Extensive manual checks — even best OCR makes mistakes on archival scans
  - Human verification of every extracted record

EXPORT DATA, 1898 AND 1904

- Hungarian Museum of Commerce
- Collection of export capable firms determined by a Hungarian institution
- Outcomes observed
  - Number of workers
  - Capacity of machinery in horsepower (hp)
  - Products offered
  - Special products and patents
  - Industry



1853. **Károlyi György, Budapest.** (V., Arany János-u. 1.) — 4 ŀP., 55 munkás.
Folyóiratok, üzleti nyomtatványok, fekete és szines litografiai térképek, építési tervek másolása és sokszorosítása, iskolai füzetek és rajztömbök.

1854. **Keppich Ferenc, Budapest.** (VI., Dávid-u. 3.) — 2 ŀP., 40 munkás.
Papirlemezből készült dobozok minden kivitelben, postadobozok.

1855. **Kiss Valdemár, Budapest.** (VIII., Üllői-út 18.) — 3 munkás.
Könyvkötészeti díszmunkák, reklámcikkek.

1856. **Kolba Mihály fiai, Diósgyőr.** (Borsod vm.) — 150 ŀP. gőzgép, 100 ŀP. turbina, 80 munkás.
Okmánypapir, értékcikkek. Különlegesség: vízjegyű biztonsági papir, merített gyártmányok.

1857. **Leszik Károly, Budapest.** (VIII., Szentkirályi-utca 6.) — 10 ŀP., 80 munkás.
Különféle egyszerű és díszes kötések. Tömeges könyvtári kötések.

1858. **Lévai Mór, Ungvár.** — 20 munkás.
Imakönyvek magyar és rutén nyelven.

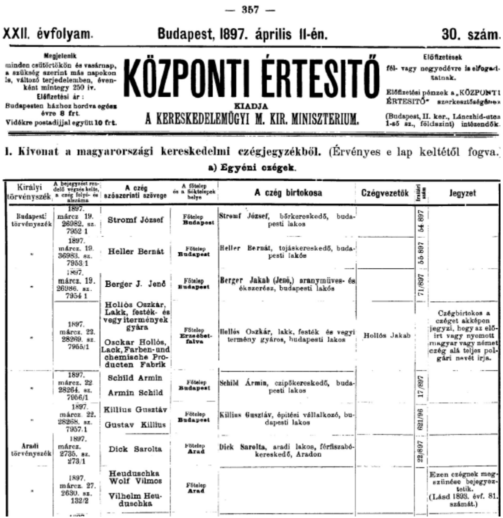
1859. **Lévay Márton könyvkötészete, vonalozóintézete és üzleti könyvek gyára, Nagyvárad.** (Fő-utca 20. Telefon: 49. Alapíttatott: 1845. év.) — 8 ŀP., 44 munkás.
Könyvkötészet, dobozgyártás, papirképkeretek, imakönyvek, képes-

DIGITIZATION PIPELINE: EXAMPLE 2: EXPORT DATA

- **Step 1: Scanning**
  - Adobe Scan for archival firm catalogues
- **Step 2: OCR**
  - Google Document AI — works efficiently with Hungarian language
  - Trained model to separate multi-column pages
- **Step 3: Parsing**
  - Challenge: firms and their attributes (HP, workers, address, product type) listed as unstructured running text
  - LLM trained via extensive few-shot examples to recognize firm boundaries and extract fields
  - Output: one row per firm with structured attributes
- **Step 4: Validation**
  - Extensive manual checks — human verification of every extracted record

CORPORATE REGISTRY, 1896-1904

- "Central Gazette" published monthy by Hungarian Ministry of Commerce
- Collection of firm births, deaths, transformations $\rightarrow$ delta-driven dataset of all firms
- Make sure we measure Paris effect on **exporting** and not **firm survival**
- We can link it to other data sources by
  - Firm name
  - Owners, managers
  - Firm location; court of registry



— 357 —

XXII. évfolyam. Budapest, 1897. április 11-én. 30. szám.

Megjelenik minden csütörtökön és vasárnap, a szükség szerint más napokon is, változó terjedelemben, évenként mintegy 250 iv. Előfizetési ár: Budapesten házhoz hordva egész évre 8 frt. Vidékre postadijjal együtt 10 frt.

KÖZPONTI ÉRTESITŐ

KIADJA A KERESKEDELEMÜGYI M. KIR. MINISZTERIUM.

Előfizetések fél- vagy negyedévre is elfogadtatnak. Előfizetési pénzek a „KÖZPONTI ÉRTESITŐ" szerkesztőségéhez (Budapest, II. ker., Lánczhid-utca 1-ső sz., földszint) intézendők.

I. Kivonat a magyarországi kereskedelmi czégjegyzékből. (Érvényes e lap keltétől fogva.)

a) Egyéni czégek.

Királyi törvényszék	A bejegyzést rendelő végzés kelte, a czég folyó- és alszáma	A czég szószerinti szövege	A főtelep és aióktelepek helye	A czég birtokosa	Czégvezetők	Iratszám	Jegyzet
Budapesti törvényszék	1897. márcz. 19. 26982. sz. 7952/1	Stromf József	Főtelep Budapest	Stromf József, bőrkereskedő, budapesti lakos		54/897	
"	1897. márcz. 19. 36983. sz. 7953/1	Heller Bernát	Főtelep Budapest	Heller Bernát, tojáskereskedő, budapesti lakós		55/897	
"	1897. márcz. 19. 26986. sz. 7954/1	Berger J. Jenő	Főtelep Budapest	Berger Jakab (Jenő,) aranymüves- és ékszerész, budapesti lakós		71/897	
"	1897. márcz. 23. 28269. sz. 7955/1	Hollós Oszkár, Lakk, festék- és vegyitermények gyára Osckar Hollós, Lack, Farben- und chemische Producten Fabrik	Főtelep Erzsébetfalva	Hollós Oszkár, lakk, festék és vegyi termény gyáros, budapesti lakos	Hollós Jakab		Czégbirtokos a czéget akképen jegyzi, hogy az előirt vagy nyomott magyar vagy német czég alá teljes polgári nevét irja.
"	1897. márcz. 22. 28264. sz. 7956/1	Schild Armin Armin Schild	Főtelep Budapest	Schild Ármin, czipőkereskedő, budapesti lakos		17/897	
"	1897. márcz. 22. 28268. sz. 7957/1	Killius Gusztáv Gustav Killius	Főtelep Budapest	Killius Gusztáv, épitési vállalkozó, budapesti lakos		621/96	
Aradi törvényszék	1897. márcz. 2735. sz. 273/1	Dick Sarolta	Főtelep Arad	Dick Sarolta, aradi lakos, férfiszabókereskedő, Aradon		22/897	
"	1897. márcz. 27. 2630. sz. 132/2	Heuduschka Wolf Vilmos Vilhelm Heuduschka	Főtelep Arad				Ezen czégnek megszünése bejegyeztetik. (Lásd 1893. évf. 81. számát.)

DIGITIZATION PIPELINE: CORPORATE REGISTRY

- Firm births/deaths/transformations 1874-1949, published monthly. **Focus**: 1895-1904
- **Step 1&2: Scanning&OCR** ✓
  - Done by Arcanum, available as PDFs ✓
- **Step 3: Parsing**
  - Challenge: table structure not captured by OCR
  - We train a Detectron2 based visual model through the LayoutParser wrapper to recognize table structure
  - Raw text to structured JSON using LLM
  - Output: list of universe of firm deaths
- **Step 4: Validation**
  - Human verification of every extracted firm death
  - Sample-based human verification of every non-death event

ENTITY RESOLUTION: LINKING FIRMS ACROSS DATASETS

- **Challenge:** same firm appears with different names across sources
  - Person names: *Kovács bunda* (fur coat maker)
  - Incorporated: *Kovács és fia bunda* (Kovács and son)
  - Renamed: *Kovács szőrme Rt* (Kovács fur joint stock co.)

- **Approach:**
  - Python text similarity packages for fuzzy matching
  - Geographic standardization via GISTA Hungarorum municipal database
  - Extensive manual checking and identification
  - Create unique ID (UID), link companies across all datasets

- **Result:** Master panel enabling balanced and unbalanced specifications

ENTITY RESOLUTION: LINKING FIRMS ACROSS DATASETS

- **Challenge:** same firm appears with different names across sources
  - Firm names: *Kovács bunda* (fur coat maker) / *Kovács és fia bunda* (Kovács and son fur coat maker) / *Kovács szőrme Rt* (Kovács fur joint stock co.)
  - OCR noise: *Schwarz* vs. *Schwarcz*, *Ferencz* vs. *Ferenez*
  - The problem is the scale: a few thousand rows of Excel spreadsheets, millions of pairing possibilities — brute force infeasible
- **Pipeline:**
  - TF-IDF blocking on character n-grams reduces to few hundred thousands candidate pairs
  - Combined score: TF-IDF cosine + token sort ratio (word-order invariant) on firm names
  - LLM validation using adress and product type (GPT-4o-mini) resolves ambiguous mid-score pairs — $\sim$\$0.30 per 20,000 pairs — human validation needed
  - Long-format output: one row per pair
- **Result:** Master panel enabling balanced and unbalanced specifications

RESULTING PANEL DATASET

- Linking firms across data sources
- Firm panel
  - 1896 Budapest trial exhibition (base). Drop firms that exited.
    - Export status before
  - 1898 Export capability before
  - 1900 Paris (treatment): 552 firms (15%)
    - Firms that exit after 1896 are interpreted as non exhibitors and thus in control group
  - 1904 Export capability after
  - 1905 Export status after
- Focus on intensive margin: firms born after 1896 not included

SUMMARY STATISTICS, 1896, 1904

Variable	N	Mean	SD	Min	Max
Treatment					
Paris exhibitor dummy	3,735	0.14	0.35	0	1
Baseline characteristics (1896, unless indicated otherwise)					
Export status	3,735	0.15	0.36	0	1
Export capability (1898)	3,735	0.25	0.43	0	1
Number of workers	1,236	175.27	781.82	1	17,165
Installed horsepower	680	222.9	978.66	1	15,700
Post-treatment (1904, unless indicated otherwise)					
Export status (1905)	3,735	0.10	0.30	0	1
Export capability	3,735	0.23	0.42	0	1

Empirical results 1: Identification

FIRM CHARACTERISTICS BY AWARD CLASS (1896)

	ln(workers)	ln(hp)	share skilled	ln(lab prod)	ln(age)
Gold medal	5.294	5.054	0.521	8.309	3.009
Silver medal	4.255	3.671	0.770	7.695	2.792
Bronze medal	3.263	2.786	0.764	7.635	2.831
Honorable mention	2.956	2.869	0.752	7.418	2.699
No award	3.527	3.401	0.874	7.352	2.831
All	3.662	3.508	0.755	7.676	2.804

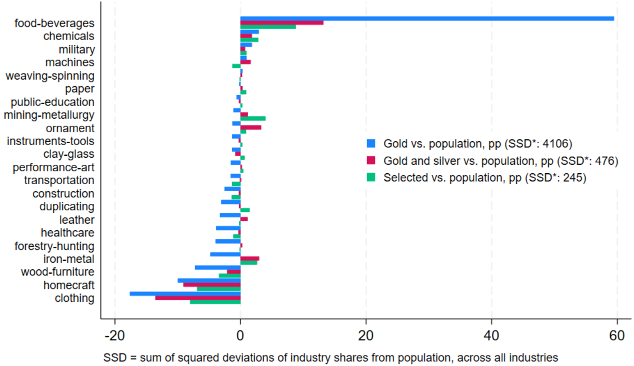
PROBABILITY OF GOING TO PARIS INCREASES IN AWARD QUALITY

Plausibility check: Probability of going to Paris increases in award quality

		Share of firms	
Award category	**Number of firms**	**Paris exhibitors**	**Non-Paris exhibitors**
Gold medal	140 (3.8%)	60.0%	40.0%
Silver medal	425 (11.4%)	36.5%	63.5%
Bronze medal	1,034 (27.7%)	16.0%	84.0%
Honorable mention	1,136 (30.4%)	6.9%	93.1%
No award	928 (26.8%)	7.0%	93.0%
Total	3,733	552	3,181
Total (%)	100%	15%	85%

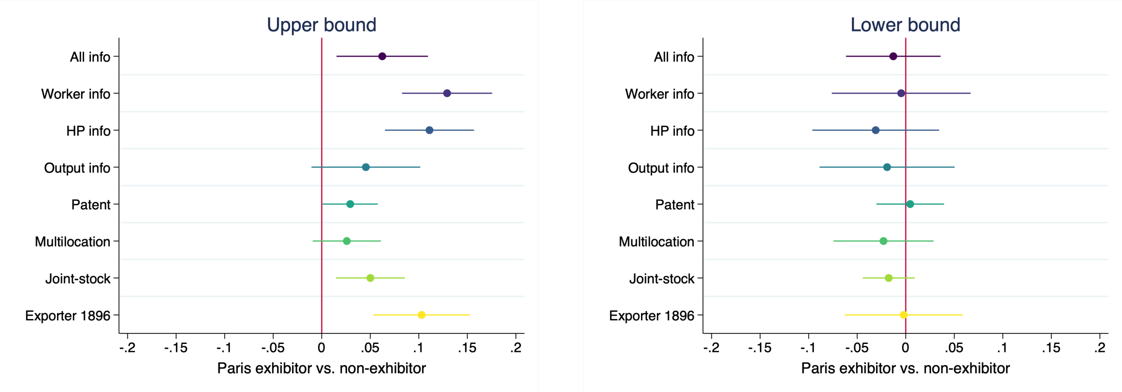
INDUSTRY BALANCE: PARIS EXHIBITORS > BUDAPEST AWARDS

Validation: Paris exhibitors are more balanced across industries than Budapest awards



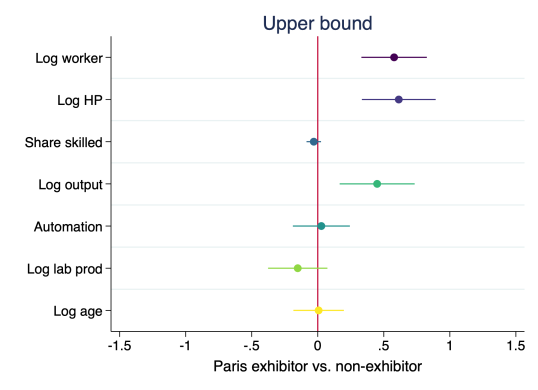
BALANCING OF PRE-TREATMENT CHARACTERISTICS 1

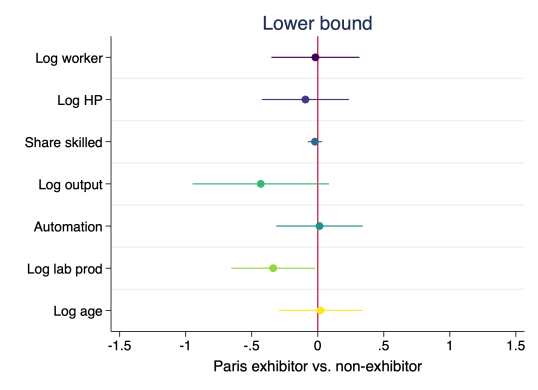
Indicator variables, available for full sample



BALANCING OF PRE-TREATMENT CHARACTERISTICS 2

Selected variables, only available for subsample





NO PRE-TRENDS IN LOWER BOUND ESTIMATION

Dep var: Δ exporting, 1896–1898

	(1)	(2) Upper bound	(3)	(4) Lower bound
Paris exhibitor	0.139***	0.121***	0.119***	0.0177
	(0.0254)	(0.0226)	(0.0268)	(0.0315)
Export status 1896	Yes	Yes	Yes	Yes
Product fixed effects	No	Yes	Yes	Yes
Bound group dummies	No	Upper	Upper	Lower
Observations	3,733	3,733	2,721	2,721
R-squared	0.018	0.291	0.293	0.289

Empirical results 2: Treatment effects

TREATMENT EFFECT: EXPORT CAPABILITY, 1904

- Estimates pass validation test: upper bound estimate $>$ lower bound estimate

Dep var: Δ export capability, 1898–1904

	(1)	(2) Upper bound	(3)	(4) Lower bound
Paris exhibitor	0.195***	0.138***	0.140***	0.0851***
	(0.0266)	(0.0228)	(0.0262)	(0.0251)
Export status 1898	Yes	Yes	Yes	Yes
Product fixed effects	No	Yes	Yes	Yes
Bound group dummies	No	Upper	Upper	Lower
Observations	3,733	3,733	2,721	2,721
R-squared	0.254	0.323	0.325	0.325

TREATMENT EFFECT: EXPORT STATUS, 1905

Dep var: Δ export status, 1896–1905

	(1)	(2) Upper bound	(3)	(4) Lower bound
Paris exhibitor	0.178***	0.101***	0.116***	0.0406*
	(0.0276)	(0.0214)	(0.0235)	(0.0239)
Export status 1896	Yes	Yes	Yes	Yes
Product fixed effects	No	Yes	Yes	Yes
Bound group dummies	No	Upper	Upper	Lower
Observations	3,733	3,733	2,721	2,721
R-squared	0.403	0.482	0.491	0.489

TREATMENT EFFECT: FIRMS GROW IN SIZE

Dep var: Δ ln(workers), 1896–1904

	(1)	(2) Upper bound	(3)	(4) Lower bound
Paris exhibitor	0.231***	0.228***	0.160***	0.161**
	(0.0531)	(0.0571)	(0.0725)	(0.0775)
ln(workers), 1896	Yes	Yes	Yes	Yes
Product fixed effects	No	Yes	Yes	Yes
Bound group dummies	No	Upper	Upper	Lower
Observations	638	638	532	532
R-squared	0.092	0.183	0.171	0.169

- Employment gains of 15–23% over 8 years $\approx$ 1.8–2.6 pp annual growth

TREATMENT EFFECT: NO SIGNIFICANT CHANGE ON MECHANIZATION

Dep var: Δ ln(horse power), 1896–1904

	(1)	(2) Upper bound	(3)	(4) Lower bound
Paris exhibitor	0.015	0.0367	0.002	-0.0717
	(0.0818)	(0.0937)	(0.120)	(0.0942)
ln(horse power), 1896	Yes	Yes	Yes	Yes
Product fixed effects	No	Yes	Yes	Yes
Bound group dummies	No	Upper	Upper	Lower
Observations	411	411	323	323
R-squared	0.058	0.198	0.216	0.219

- Firms expanded through labor rather than capital-intensive upgrades

SUMMARY OF MAIN RESULTS

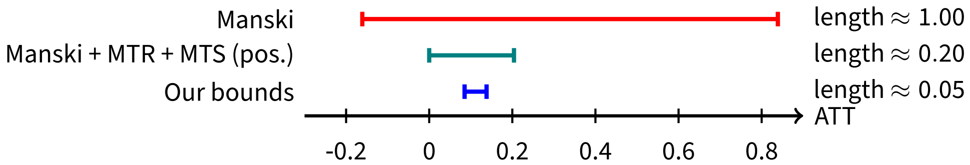
Outcome	Lower bound	Upper bound
Export capability (1904)	+8.5 pp ***	+13.8 pp ***
Export status (1905)	+4.1 pp *	+10.1 pp ***
Employment growth	+16.1% **	+22.8% ***
Mechanization (HP)	n.s.	n.s.

- **All main outcomes pass validation test:** $\beta_{\text{upper}} \geq \beta_{\text{lower}}$
- Trade fair participation has a positive causal effect on exports and employment
- Firms expanded through labor, not through mechanization

Manski

COMPARISON WITH MANSKI BOUNDS

Δ export capability, 1898–1904, controlling for export capability 1898; all applied to ATT



- **Manski bounds:** $[-0.16, 0.84]$
  - Conditioning on previous export status, export shares are between 0 and 1
- **Manski + MTR + MTS (pos.):** $[0.00, 0.204]$
  - Lower bound: set to 0 due to monotone treatment response (MTR)
  - Upper bound: assuming monotone (positive) treatment selection (MTS, “naive” estimate)
- **Our bounds:** $[0.0851, 0.138]$

Competition and heterogeneity

LARGE VARIATION IN COMPETITION AND BUYERS

Hungarian competitors
(firms in same Paris subcategory)

- Mean: 44, range: 3–157
- Few: Electrochemistry, Silks
- Many: Wines, Food industries

International buyers
(exhibitors in buyer product categories)

- Mean: 3,149, range: 92–13,874
- Few: Hunting weapons, Musical instruments
- Many: Flour products, Mining

- Both measures vary substantially across the 72 Paris subcategories with Hungarian firms
- Measured in logs in regressions

HETEROGENEOUS EFFECTS: REGRESSION SPECIFICATION

Extend the bounding regression with interaction terms:

$$\Delta y_i = \beta \, \text{exhibitor}_i + \gamma \, \text{exhibitor}_i \times \ln(\text{competition}_k) + \delta \, \text{exhibitor}_i \times \ln(\text{buyers}_k) + \text{BoundGroupFE}_j +$$

- γ: effect of domestic competition on treatment effect
  - Expected sign ambiguous: congestion vs. attracting more buyers
- δ: effect of potential buyer pool on treatment effect
  - Expected positive: thicker market $\Rightarrow$ better matching
- Lower bound estimates; product fixed effects included

COMPETITION REDUCES EFFECT ON EXPORTING

Dep var: Δ export capability, 1898–1904. Lower bound estimates.

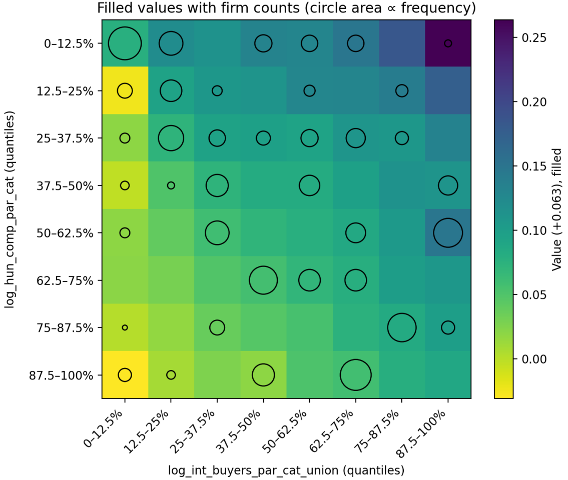
	(1)	(2)
Paris exhibitor * ln(Hungarian competition)	-0.0418*	-0.0428*
	(0.0249)	(0.0250)
Paris exhibitor * ln(number of buyers)	0.0414**	
	(0.0167)	
Paris exhibitor * ln(number of international buyers)		0.0434**
		(0.0175)
Product effects	Yes	Yes
Bound group dummies	Lower	Lower
Sample	Lower	Lower
Observations	2,721	2,721
R-squared	0.487	0.487

ECONOMIC SIGNIFICANCE OF HETEROGENEITY

Quartile comparisons:

- Competition: p75 vs. p25 difference = 1.34 log units ⇒ 5.7pp lower export probability
  - **66%** of average TE
- Buyers: p75 vs. p25 difference = 0.87 log units ⇒ 3.7pp higher export probability
  - **43%** of average TE

Both effects are economically large relative to the 8.6pp ATE



Heatmap: TE by competition × buyers

WHAT DO WE LEARN ABOUT HOW TRADE FAIRS WORK?

- **Trade fairs reduce matching frictions** by pooling suitable “candidates”
- **Two competing forces:**
  - Matching benefits: more potential buyers $\Rightarrow$ higher chance of finding a partner
  - Congestion effects: more domestic competitors $\Rightarrow$ lower chance of standing out
- **Our findings:**
  - No one loses from participation on average
  - But: more competitors at the fair $\Rightarrow$ smaller conversion to exporter
  - More international buyers $\Rightarrow$ larger conversion to exporter
- **Policy implication:** Trade fair design matters — composition of participants affects outcomes

CONCLUSION

- **Effect of trade fair participation**
  - Increased export probability by at least 4–10% by 1904/1905
  - Led to 15–23% employment growth, but no significant increase in mechanization
  - Export probability declines with more intense domestic competition but increases with access to potential buyers
- **Methodological contribution**
  - ‘Causal bands’: in ‘fuzzy RDD’ setting, choose control group above and below treatment group to bound the estimate
  - Compared to matching, which just focuses on ‘close’ observations, we exploit additional information on whether ‘close’ is ‘close from above’ versus ‘close from below’ to bound the estimates
  - Significantly tighter than standard Manski bounds

Comments welcome!

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LE PALAIS DES ILLUSIONS

C'est une vaste salle hexagonale revêtue, sur ses six côtés, d'immenses glaces de Saint-Gobain, et couverte par un plafond doré, sculpté dans le style mauresque par M. Alméras. Une série de lampes électriques, installées avec le plus grand art par MM. Picou et Martine, éclairent de tons variés et changeants des colonnes, des appliques, des girandoles qui se reflètent à l'infini, et donnent l'illusion d'une féerique mosquée illuminée de cent mille feux.

EXPORT CAPABILITY: DEFINITION

- Firm survey by the Hungarian Museum of Commerce 1904
- **Objective:** all factories "which only lacked a foreign order to become export companies"; not an "export," but an "export-capable" directory of domestic manufacturers
- **Export capability:** industrial plants that exceed local consumption demands; are ready to ship abroad and engage with foreign importers
- Starting point: regular survey of the Hungarian industry 1899
- Updated with new firms and firms which ceased operations
- Information about export capability through
    - Sending questionnaires to more than 4,200 domestic manufacturing plants; 3 reminder letters
    - Business associations
    - Advertisements in more than 200 domestic newspapers

back

MANSKI BOUNDS: IDEA AND MOTIVATION

Goal: Learn about the average treatment effect (ATE)

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)]$$

when we only observe

$$Y = DY(1) + (1 - D)Y(0), \quad D \in \{0, 1\},$$

and we **do not** assume random assignment, selection-on-observables, instruments, etc.

Manski's idea:

- Work under extremely weak assumptions (e.g. only that Y is bounded: $Y \in [0, 1]$).
- We cannot point-identify ATE, but we can find **sharp bounds**:

$$\underline{\text{ATE}} \leq \text{ATE} \leq \overline{\text{ATE}}.$$

- These are the *Manski bounds* (or *worst-case bounds*).

DERIVING MANSKI'S WORST-CASE ATE BOUNDS

The ATE can be written as

$$\text{ATE} = p_1\mu_1 + p_0\theta_1 - p_1\theta_0 - p_0\mu_0,$$

where $\mu_1 = \mathbb{E}[Y \mid D = 1]$, $\mu_0 = \mathbb{E}[Y \mid D = 0]$, and $\theta_1 = \mathbb{E}[Y(1) \mid D = 0]$, $\theta_0 = \mathbb{E}[Y(0) \mid D = 1]$ are unknown.

Under only $Y \in [0, 1]$: $0 \leq \theta_1, \theta_0 \leq 1$.

Worst-case bounds:

$$\begin{aligned} \underline{\text{ATE}} &= p_1(\mu_1 - 1) - p_0\mu_0, \\ \overline{\text{ATE}} &= p_1\mu_1 + p_0(1 - \mu_0). \end{aligned}$$

TIGHTENING THE BOUNDS

Tightening the bounds:

- *Monotone Treatment Response (MTR):* $Y(1) \geq Y(0)$ for all units
- *Monotone Treatment Selection (MTS):* treated units have weakly higher potential outcomes
- Instrumental variables with partial identification

Each assumption shrinks $[\underline{\text{ATE}}, \overline{\text{ATE}}]$, trading robustness for informativeness.

Our approach: Award-based bounds deliver much tighter intervals by exploiting institutional features of the selection process.

REFERENCES

Alfaro, Laura, Cathy Ge Bao, Maggie X. Chen, Junjie Hong, and Claudia Steinwender, "Omnia Juncta in Uno: Foreign Powers and Trademark Protection in Shanghai's Concession Era," NBER Working Paper 29721 2022.

Allen, Treb, "Information frictions in trade," *Econometrica*, 2014, *82* (6), 2041–2083.

Arkolakis, Costas, Federico Huneeus, and Yuhei Miyauchi, "Production Network Formation, Trade, and Welfare," 2025.

Atkin, David, Amit K. Khandelwal, and Adam Osman, "Exporting and Firm Performance: Evidence from a Randomized Experiment," *The Quarterly Journal of Economics*, 2017, *132* (2), 551–615.

Benguria, Felipe, "The matching and sorting of exporting and importing firms: theory and evidence," *Journal of International Economics*, 2021, *131*, 103430.

REFERENCES

Bernard, Andrew B and J Bradford Jensen, "Why some firms export," *Review of economics and Statistics*, 2004, *86* (2), 561–569.

— **, Andreas Moxnes, and Yukiko U Saito**, "Production networks, geography, and firm performance," *Journal of Political Economy*, 2019, *127* (2), 639–688.

Biesebroeck, Johannes Van, Emily Yu, and Shenjie Chen, "The impact of trade promotion services on Canadian exporter performance," *Canadian Journal of Economics/Revue canadienne d'économique*, 2015, *48* (4), 1481–1512.

Blundell, Richard, Amanda Gosling, Hidehiko Ichimura, and Costas Meghir, "Changes in the Distribution of Male and Female Wages Accounting for Employment Composition Using Bounds," *Econometrica*, 2007, *75* (2), 323–363.

Broocks, Annette and Johannes Van Biesebroeck, "The impact of export promotion on export market entry," *Journal of International Economics*, 2017, *107*, 19–33.

REFERENCES

Buus, Magnus T, Jakob R Munch, Joel Rodrigue, and Georg Schaur, "Do Export Support Programs Affect Prices, Quality, Markups, and Marginal Costs? Evidence From a Natural Policy Experiment," *Review of Economics and Statistics*, 2025, *107* (1), 172–187.

Cai, Jing, Wei Lin, and Adam Szeidl, "Firm-to-Firm Referrals," 2024.

Carballo, Jerónimo and Christian Volpe Martincus, "Is export promotion effective in developing countries? Firm-level evidence on the intensive and the extensive margins of exports," *Journal of International Economics*, 2008, *76* (1), 89–106.

Demir, Banu, Ana Cecilia Fieler, Daniel Yi Xu, and Kelly Kaili Yang, "O-ring production networks," *Journal of Political Economy*, 2024, *132* (1), 200–247.

Donaldson, Dave and Richard Hornbeck, "Railroads and American economic growth: A "market access" approach," *The Quarterly Journal of Economics*, 2016, *131* (2), 799–858.

Eaton, Jonathan, David Jinkins, James R Tybout, and Daniel Xu, "Two-sided search in international markets," Technical Report, National Bureau of Economic Research 2022.

REFERENCES

Görg, Holger, Michael Henry, and Eric Strobl, "Grant support and exporting activity," *The review of economics and statistics*, 2008, *90* (1), 168–174.

Hanlon, W Walker, "The persistent effect of temporary input cost advantages in shipbuilding, 1850 to 1911," *Journal of the European Economic Association*, 2020, *18* (6), 3173–3209.

Hiller, Sanne, "Does export promotion work in Denmark? Evidence from a matching approach," 2012.

Ho, Kate and Adam M Rosen, "Partial identification in applied research: benefits and challenges," *Advances in Economics and Econometrics*, 2017, *2*, 307–359.

Jensen, Robert, "The digital provide: Information (technology), market performance, and welfare in the South Indian fisheries sector," *The Quarterly Journal of Economics*, 2007, *122* (3), 879–924.

— **and Nolan H Miller**, "Market integration, demand, and the growth of firms: Evidence from a natural experiment in India," *American Economic Review*, 2018, *108* (12), 3583–3625.

REFERENCES

Juhász, Réka and Claudia Steinwender, "Spinning the web: The impact of ICT on trade in intermediates and technology diffusion," Technical Report, National Bureau of Economic Research 2018.

Juhász, Réka, "Temporary Protection and Technology Adoption: Evidence from the Napoleonic Blockade," *American Economic Review, 108 (11): 3339-76.*, 2018.

Lee, David S, "Training, wages, and sample selection: Estimating sharp bounds on treatment effects," *The Review of Economic Studies*, 2009, *76* (3), 1071–1102.

Manski, Charles F, *Partial identification of probability distributions*, Springer, 2003.

Manski, Charles F. and John V. Pepper, "Monotone Instrumental Variables: With an Application to the Returns to Schooling," *Econometrica*, 2000, *68* (4), 997–1010.

Martincus, Christian Volpe and Jerónimo Carballo, "Beyond the average effects: The distributional impacts of export promotion programs in developing countries," *Journal of Development Economics*, 2010, *92* (2), 201–214.

REFERENCES

Miyauchi, Yuhei, "Matching and Agglomeration: Theory and Evidence from Japanese Firm-to-Firm Trade," *Econometrica*, 2024, *92* (6), 1869–1905.

Moser, Petra, "How Do Patent Laws Influence Innovation? Evidence from Nineteenth-Century World's Fairs," *American Economic Review*, September 2005, *95* (4), 1214–1236.

__ , "Innovation without Patents: Evidence from World's Fairs," *The journal of law economics*, 2 2012, *55* (1), 43–74.

Munch, Jakob and Georg Schaur, "The effect of export promotion on firm-level performance," *American Economic Journal: Economic Policy*, 2018, *10* (1), 357–387.

Munch, J.R., M. Buus, J. Rodrigue, and G. Schaur, "Do Export Support Programs affect Prices, Quality, Markups and Marginal Costs? Evidence from a Natural Policy Experiment," *Review of Economics and Statistics*, forthcoming.

Rauch, James E, "Business and social networks in international trade," *Journal of economic literature*, 2001, *39* (4), 1177–1203.

REFERENCES

— **and Vitor Trindade**, "Ethnic Chinese networks in international trade," *Review of Economics and Statistics*, 2002, *84* (1), 116–130.

Romero, Dario A, "An Empire Lost: Spanish Industry and the Effect of Colonial Markets on Trade and Innovation," 2023.

Schminke, Annette and Johannes Van Biesebroeck, "Using export market performance to evaluate regional preferential policies in China," *Review of World Economics*, 2013, *149* (2), 343–367.

Sugita, Yoichi, Kensuke Teshima, and Enrique Seira, "Assortative matching of exporters and importers," *Review of Economics and Statistics*, 2023, *105* (6), 1544–1561.

Tamer, Elie, "Partial identification in econometrics," *Annual Review of Economics*, 2010, *2* (1), 167–195.

Verhoogen, Eric A, "Trade, quality upgrading, and wage inequality in the Mexican manufacturing sector," *The quarterly journal of economics*, 2008, *123* (2), 489–530.